

SMART PARKING LOCK

Product Requirements Document — Hardware Implementation Scope

Sections 1–7 | v4

1. Product Dimensions & Packaging

Defines the physical footprint and packaging requirements for safe transport, storage, and installation across residential, commercial, and government deployments.

Key Requirements

- Compact footprint compatible with standard parking-bay widths
- Low profile when lowered (open) to minimize tripping / vehicle-clearance risk
- Manageable weight for 1–2 person handling and installation
- Standardized, protective packaging for logistics and storage

Design Requirements

- Final L × W × H to be confirmed after CAD
- Packaging to withstand transit shock, vibration, and moisture
- Clear handling / orientation labeling on packaging

Validation

- Dimensional / tolerance check
- Drop test
- Transit vibration test

Dimensions: 46 × 46 × 9 cm; Weight: ~8 kg-(-aprox acc. Parkobot)

2. Structural Design

Defines the mechanical architecture of the product — modular, durable, and serviceable across residential, commercial, and government deployments.

Key Modules

- Barrier Module — vehicle blocking, impact absorption, branding
- Drive Module — motor, gearbox, coupling, pivot shaft
- Chassis Module — structural rigidity, load distribution
- Housing Module — environmental & mechanical protection
- Mounting Module — anchoring and ground interface

Construction

- Barrier: UV-resistant rubber/PU cover over a segmented aluminium or glass-filled nylon spine, with impact-absorbing foam and an internal steel safety cable
- Chassis: 2–3 mm powder-coated galvanized steel with dedicated mounts for motor, battery, lock, and PCB
- Housing: sealed, UV-stabilised enclosure with replaceable gasket (IP67 target)
- Pivot: stainless-steel shaft with replaceable bushes and mechanical end stops

Design Targets

- Design life ≥ 5 years | $\geq 100,000$ operating cycles | Safety factor ≥ 2
- Corrosion & UV resistant, IP67 target, no exposed sharp edges or pinch points

3. Locking Mechanism

Core design rule: the motor moves the barrier; a separate steel lock secures it. This separation keeps holding loads off the motor and gearbox.

Proposed Components

- 12V DC geared motor
- Spring-loaded, hardened-steel locking pin (deadbolt)
- Solenoid or micro-servo for pin release
- Two Hall sensors (raised / lowered position) + one lock-engagement sensor
- Motor-current sensor for jam / forced-movement detection
- Concealed manual key release for power-failure access

Operating Sequence

Authorized command $\rightarrow$ lock disengages $\rightarrow$ sensor confirms $\rightarrow$ motor moves barrier $\rightarrow$ position sensor confirms $\rightarrow$ mechanical lock engages $\rightarrow$ secure status shown.

Tamper / Overload Response

- Forced movement is absorbed by the steel pin and chassis, not the motor
- Buzzer, LED, and app/event-log notification trigger on unexpected movement
- On heavy impact, the flexible arch yields in a controlled manner while lock, motor, and electronics stay protected
- Lock remains mechanically engaged during power loss; no continuous power required

Terminology note: raised barrier = Locked/Protected Position; lowered barrier = Open Position. Keep this convention consistent across CAD, firmware, and testing.

self-locking worm gear-

4. Motion System

Delivers smooth, controlled, and repeatable barrier motion, distinct from the locking/security function covered in Section 3.

Key Requirements

- Smooth start/stop motion with no abrupt jerks
- Defined, repeatable open/close cycle time
- Real-time obstruction / overload detection during motion
- Stable performance across temperature and load variation

Components

- DC geared motor
- Motor driver / controller
- Transmission (gearbox / coupling)
- Position feedback sensors (Hall / encoder)

Design Requirements

- Cycle time target: ≤ 5 seconds open/close (to be validated)
- Auto-stop and reverse on obstruction detection
- Manual motion override during power failure

Validation

- Cycle time test
- Repeatability test
- Obstruction detection test

5. Mounting System

A secure, vandal-resistant mounting system enabling fast installation and long-term stability on concrete, asphalt, and paver surfaces.

Key Components

- Reinforced steel base plate + chemical/expansion anchor bolts
- Mounting bracket with anti-tamper fasteners
- Rubber sealing gasket and alignment template

Design Requirements

- Minimum 4 mounting points; hidden/protected, corrosion-resistant fasteners
- Anti-pry base geometry; waterproof seal between housing and ground

Design principle: the mounting system must be stronger than the barrier assembly, so external forces reach the ground without loosening the device.

6. Battery & Power Housing

A safe, weather-resistant, serviceable compartment for the battery and power components, supporting multiple charging options.

Power Configurations

- Battery only | Battery + Solar | Battery + AC Adapter

Key Requirements

- IP67-ready sealed compartment with secure mounts for battery and BMS
- Ventilation for heat dissipation; waterproof cable entry points
- Modular battery tray for quick, tool-assisted replacement
- Battery physically isolated from the motor and gearbox

(Battery life: Up to 4–6 months (typical use))--parkobot

7. Serviceability

The product must allow quick inspection, repair, and part replacement in the field — without full disassembly — backed by a predictive, modular, remote-first maintenance strategy that keeps lifetime ownership cost low.

Maintenance Levels

Level	Description
L1	Cleaning & Inspection
L2	Battery & Barrier Replacement
L3	Motor, Lock & PCB Replacement
L4	Factory Repair (Major Damage)

Field-Replaceable Components & Typical Service Time

Component	Replace Time
Battery Pack	5 min
Barrier Module	10 min
Motor & Gearbox	15 min
Main PCB	10 min
Communication Module	5 min
Sensors	5 min

- Also field-serviceable: charging module, seals & gaskets

Warranty

- 1-year standard warranty covering manufacturing defects
- Free OTA firmware updates and free mobile app updates

Predictive Maintenance (Key USP)

- Continuous monitoring: battery health, motor current, lock status, sensor status, communication status, tamper events, temperature
- On issue detection: alert to owner + dashboard, service ticket auto-generated

AMC Plans

Plan	Residential	Commercial	Government
Basic	Free (1-yr warranty)	Included	Included
Standard	₹**/year	/unit/year	Custom SLA
Premium	₹, ***/year	/unit/year	Custom SLA + 24×7 support

Remote Maintenance

- OTA firmware updates, device diagnostics, configuration changes, log collection, health monitoring, remote restart — reduces technician visits

Preventive Maintenance Schedule

Interval	Activity
Every 6 months	Cleaning, bolt inspection, functional test
Every 12 months	Battery health check, seal inspection, firmware update
Every 3–5 years	Battery replacement (usage-dependent)

Service Dashboard

- Device health, battery %, last online, signal strength, motor & lock status, error codes
- Service history, warranty status, AMC status

Design Targets

- Average annual maintenance cost: ₹200–500 per unit
- Average service time: <20 minutes
- Remote issue resolution: >80%
- Device uptime: >99%

USP: Predictive maintenance, remote diagnostics, OTA updates, and modular components — designed to minimize downtime and reduce lifetime ownership cost.

Conclusion — What This Document Covers

This document defines the complete Hardware Implementation Scope for the Smart Parking Lock across seven core areas:

- Product Dimensions & Packaging — footprint, weight, and transit/packaging requirements
- Structural Design — mechanical architecture, modules, materials, and design targets
- Locking Mechanism — motor + steel-lock separation, components, and tamper response
- Motion System — drive components, cycle time, and obstruction handling
- Mounting System — installation, anchoring, and validation requirements
- Battery & Power Housing — power configurations and housing requirements
- Serviceability — maintenance levels, modular replacement, warranty, predictive maintenance, AMC plans, remote maintenance, and service targets

Electronics Implementation Scope, Project Timeline, and Research Items are covered separately and will be added in a follow-up version.